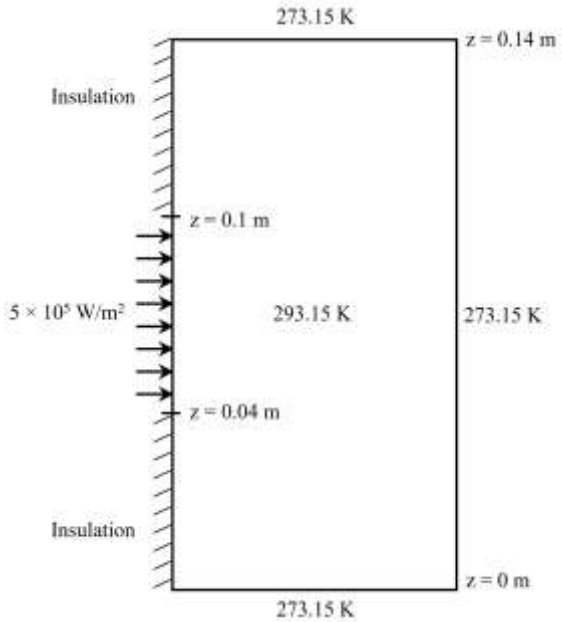
UNIVERSITY OF DHAKA
Department of Applied Mathematics
Fourth Year B.S. in Applied Mathematics, Academic Session: 2023 – 2024
Course No: **AMTH 450**, Course Title: **MATH LAB IV (Application Software)**
Assignment – 7: Solving real-life problems using COMSOL Multiphysics

Instruction: Develop simulation code using the COMSOL Multiphysics Interface to solve the following problems and visualize the corresponding results.

Full Name: **Roll:** **Group:**

Q1. The modeling domain describes the cross section of a 2D solid with dimensions $0.08m \times 0.14m$, as shown in the accompanying figure.



(a) Create the prescribed model geometry in a 2D axisymmetric space domain.
(b) Use the physics interface to implement the governing equations:

$$\rho C_p \frac{\partial T}{\partial t} + \nabla \cdot \left(-k \nabla T\right) = Q$$

, where $k = 52 W/(m.K), \rho = 7850 kg/m^3, C_p = 460 J/(kg.K), Q = 0$.

(c) Implement the initial and boundary conditions according to the specifications in the accompanying figure, and generate the mesh networks accordingly.
(d) Solve the above problems described above under stationary case, and transient case $(t = 60s)$, and visualize the temperature distributions as well as the isothermal contours.
(e) Compute the temperature at the specified locations: A (0.04 m, 0.04 m), B (0.06 m, 0.07 m), and C (0.06 m, 0.1 m). Also, plot the temperature profiles along the lines at axial positions of 0.05 m, 0.08, and 0.1 m, as well as at radial positions of 0.03 m, 0.05 m, and 0.07 m.

Q2. Consider an internal carotid artery (ICA) occupied with blood having a constant density of 1080 kg m^{-3}, and a constant viscosity of 0.00325 Pa.s. The ICA has a diameter of 5.57 mm, and a length of 53 mm. An ellipsoid represented the carotid sinus (CS) with a major axis of 4.6 mm and a minor axis of 1.55 mm. The CS has the same axis as the ICA walls and its center is located at a distance of 19 mm (bottom wall) and 38 mm (top wall) from the initial position of the ICA.

(a) Draw a 2D geometry model of the ICA sinus using COMSOL Multiphysics tool.

(b) Adopt the governing equations of the hemodynamics used in physics interface that are stated as follows:

$$\nabla \cdot \boldsymbol{u} = 0$$

$$\rho \frac{\partial \boldsymbol{u}}{\partial t} + \rho(\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \mu \nabla^2 \boldsymbol{u}$$

(c) Solve the above problem with pulsatile flow at the inlet defined by $\boldsymbol{u} = -U_0 \boldsymbol{n}$, where

$$U_0 = u_{\max} \cdot f(t);\ f(t) = \begin{cases} 0.8\sin(2.857\pi t), & 0 \le t \le 0.35 \\ 0.6\sin(2.22\pi(t-0.35)), & 0.35 \le t \le 0.8 \end{cases},\ u_{\max} = 0.5\, ms^{-1}$$

, and a simpler uniform gauge pressure of zero at the outlet. No-slip boundary condition is specified at the ICA wall.

(d) After post-processing to find **(i)** the velocity and pressure contours inside the ICA, **(ii)** the wall shear stress (WSS) along the ICA walls, and **(iii)** the velocity profiles while creating the lines at the positions of 8 mm, 15 mm, 19 mm, and 38 mm. Also, apply parametric sweep approach for the velocity of $u_{max} = 0.25, 0.45, 0.65, 0.85\, ms^{-1}$ to find the results of **parts (i), (ii), and (iii)**.

Q3. The artificial porous model consists of square pores with a side length of 0.08 mm, positioned at (0.01 mm, 0.01 mm), each containing a circular arc of radius 0.016 mm. Construct an array of these pore structures with a size of 8×9, using a uniform displacement of (0.1 mm, 0.1 mm) between adjacent pores.

(a) Develop a 2D geometric model representing the artificial porous structures with a side length of 0.8 mm, positioned at the origin.

(b) Apply the physics interface to introduce the following time-dependent model equation:

$$\frac{\partial c}{\partial t} + \nabla \cdot \left(-D \nabla c\right) = R,$$

(c) Solve the problem by applying a concentration boundary condition (i.e., $c = c_0$) on the left vertical edge, a flux boundary condition (i.e., $(-D\nabla c) \cdot \boldsymbol{n} = k_m (c - c_1)$) on the right vertical side, and insulating boundary conditions (i.e., $(-D\nabla c) \cdot \boldsymbol{n} = 0$) on all other boundaries. The initial conditions is set as $c(t=0) = c_0 e^{-a\left(\frac{x}{0.4}\right)^2}$. Take $c_0 = 3 mol/m^3, a = 1000, D = 1 \times 10^{-5} m^2/s, k_m = 5 m/s, c_1 = 0, R = 0$.

(d) Visualize the surface concentrations and streamline at t = 15 s, and compute the concentrations at the specified locations: A (0.1 mm, 0.4 mm), B (0.5 mm, 0.6 mm), and C (0.7 mm, 0.3 mm) at t = 20s. Also, plot the temperature profiles along the vertical lines at positions of x = 0.1 mm, 0.2 mm, 0.7 mm, and 0.8 mm.

Q4. The transport equations can be described by the following equations:

$$\nabla \cdot \boldsymbol{u} = 0$$

$$\rho\left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\rho\left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

The dimensionless equations are stated as below:

$$\nabla \cdot \boldsymbol{u} = 0$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$$

(a) Generate 2D geometries based on a unit aspect ratio for the following shapes: **(i)** a right-angled triangle, **(ii)** an isosceles triangle, and **(iii)** an equilateral triangle, each filled with water.

(b) Implement the above dimensionless equations for each case with **(i)** no slip wall boundary conditions and **(ii)** slip wall boundary conditions with initial velocity, **u** = (1, 1) and $Re = 10^3$.

(c) Solve the problems and visualize the velocity and pressure distributions for $Re = 10^4$, 10^5, and 10^7.

Q5. Consider a square cell with a unit aspect ratio filled with water, where the bottom wall is maintained at a higher temperature (hot) and the top wall at a lower temperature (cold). The vertical side walls are thermally insulated. The fluid motion and heat transfer within the cell are governed by the buoyancy-driven convection. This physical process is described by the following set of dimensionless equations:

$$\nabla \cdot \boldsymbol{u} = 0$$

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$$

$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\mathrm{Re}}\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + \frac{Gr}{\mathrm{Re}^2}T$$

$$\frac{\partial T}{\partial t} + u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y} = \frac{1}{\mathrm{Pr}\,\mathrm{Re}}\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)$$

Solve the above the problem with (i) no slip conditions on all walls and (ii) free slip boundary conditions. The dimensionless temperature is set as $T = 1$ at the bottom boundary and $T = 0$ at the top boundary. Choose $Gr = 10^2$, 10^4, 10^6, $Re = 10^3$, 10^4, 10^5, $Pr = 6.9$ to show the temperature distributions and velocity distributions at time $t = 30$s with initial velocity (i) **u** = (0, 0) and (ii) **u** = (1, 1). Also, plot the Nusselt number profiles along the hot wall for all cases.

Q6. A fluid of viscosity μ_1, injected at a uniform velocity U, displaces another fluid of viscosity $\mu_2 (> \mu_1)$ in two-dimensional homogeneous porous media having constant permeability κ and constant porosity ε as illustrated in the accompanying figure. Concentration of a solute in the solvent is denoted by c. Without loss of generality, $c = 0\, mol/m^3$ for fluid of viscosity μ_1 and $c = c_2\, mol/m^3$ for fluid of viscosity μ_2. As the fluids are assumed miscible in nature, so a thin transition zone between the fluids immediately created and we assume it has thickness δ at the beginning of our simulations, as indicated in the figure. Assume, domain of our problem is $L \times H$ and x_0 is the interface position. Fluid motion in a porous media is given by the Darcy's law:

$$\nabla \cdot \boldsymbol{u} = 0, \boldsymbol{u} = -\frac{\kappa}{\mu(c)}\nabla p$$

The boundary conditions are defined as $\boldsymbol{u} = (U, 0)$ at $x = 0$, and $p = 0$ at $x = L$. $\frac{\partial u}{\partial x} = 0$, $v = 0$ at $y = 0$ and $y = H$.

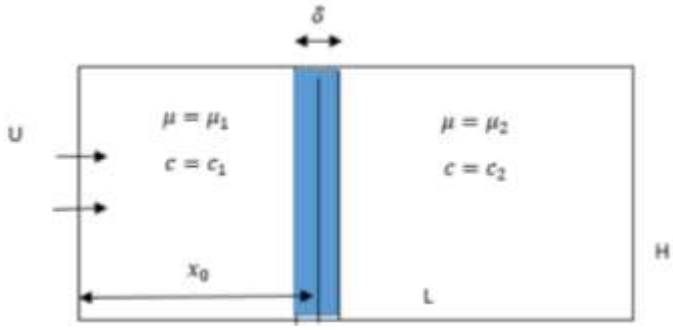


Figure: Physical Model

Transport of the solute that controls the dynamic viscosity of the fluids is described by a convection-diffusion equation:

$$\varepsilon \frac{\partial c}{\partial t} + \boldsymbol{u} \nabla c = \varepsilon D \nabla^2 c$$

where D is the diffusion coefficient. The boundary conditions associated with this equation are, $c = 0$ at $x = 0$, $\frac{\partial c}{\partial x} = 0$ at $x = L$, $\frac{\partial c}{\partial x} = 0$ at $y = 0$ and $y = H$. A transition region is of thickness δ exist between the two fluids. The initial condition for concentration is given by $c(x, y, t = 0) = \frac{c_2}{2}\left(1 + erf\left(\frac{x - x_0}{\delta}\right)\right)$ and $\mu(c) = \mu_1 e^{\frac{Rc}{c_2}}$. Some important parameter values are $L = 0.08m$, $H = 0.02m$, $R = 2$, $U = 1$ mm/s, $\mu_1 = 1$ $mPa.s$, $\rho = 1000$ $kg/m3$, $D = 4\times10^{-8}$ m^2/s, $\kappa = 10^{-6}$ m^2, $x_0 = 0.01$ m, $\delta = 10^{-4}$ m. Use different porosity values $\varepsilon = 0.1, 0.5, 0.75$ and simulate for enough time so that you can see the viscous fingering in the concentration profile.

The End